

Python For Finance Professionals

Back Testing

Objectives

By the end of this session, participants will be able to:

What is Back Testing

Objectives

Backtesting

The process of evaluating a trading strategy by applying it to historical market data to simulate how it would have performed.

Why Backtest?

It helps answer the question: **“Would this strategy have worked in the past?”**

Provides quantitative insights into performance before risking real capital.

Allows developers to debug logic and refine strategy rules.

Core Concepts

Trading Signals: Buy, sell, or hold decisions generated by rules (e.g., moving average crossover).

Historical Data: Past prices, volumes, and indicators are used as the input.

Execution Logic: Positions are simulated using signals and market returns.

Performance Metrics: Returns, volatility, Sharpe ratio, drawdown, etc.

Example Strategies Covered

Sell In May

Sell in May and go away, always remember to buy back in November

Moving Average Crossover

Enters and exits positions based on short-term vs long-term trend signals.

Bollinger Bands

Uses price deviation from a moving average to identify overbought/oversold conditions.

Completely Random

Just randomly pick a position

Typical Back Test Workflow

Load Historical Data

- Daily or intraday prices (open, high, low, close, volume).
- Clean and preprocess data if necessary.

Generate Signals

- Apply trading logic (e.g., crossover, Bollinger Bands).
- Output a signal: +1 (Long), -1 (Short), or 0 (Cash/Neutral).

Simulate Positions

- Shift signals forward by 1 day to reflect real-world execution.
- Track position over time (held, exited, flipped).

Calculate Strategy Returns

- Multiply market returns by lagged positions.
- Use log returns or percent returns as needed.

Evaluate Performance

- Cumulative return
- Sharpe ratio
- Drawdown
- Volatility
- Compare against the market or a benchmark

Visualise Results

- Plot price and signals.
- Plot cumulative returns of strategy vs market.

Limitations of Backtesting

Overfitting

- Tuning strategy parameters to fit historical data too well.
- May lead to high past performance but poor results in live trading.

Lookahead Bias

- *Using future data (e.g. today's close to trigger a trade today) without realizing it.*
- Leads to unrealistic performance — impossible to reproduce in real time.

Survivorship Bias

- Using datasets that exclude delisted or failed companies.
- Can artificially inflate returns by only including “winners.”

Ignoring Execution Realities

- No accounting for slippage, transaction costs, spreads, or market impact.
- Strategy may look profitable on paper but not in real markets.

False Confidence

- Good backtest results $\neq$ good strategy.
- Without sound reasoning or market intuition, strong backtest performance may be misleading.

Lack of Robustness

- Strategy may only work in a specific historical period or market condition.
- Doesn't generalize across assets or timeframes.

Sell in May

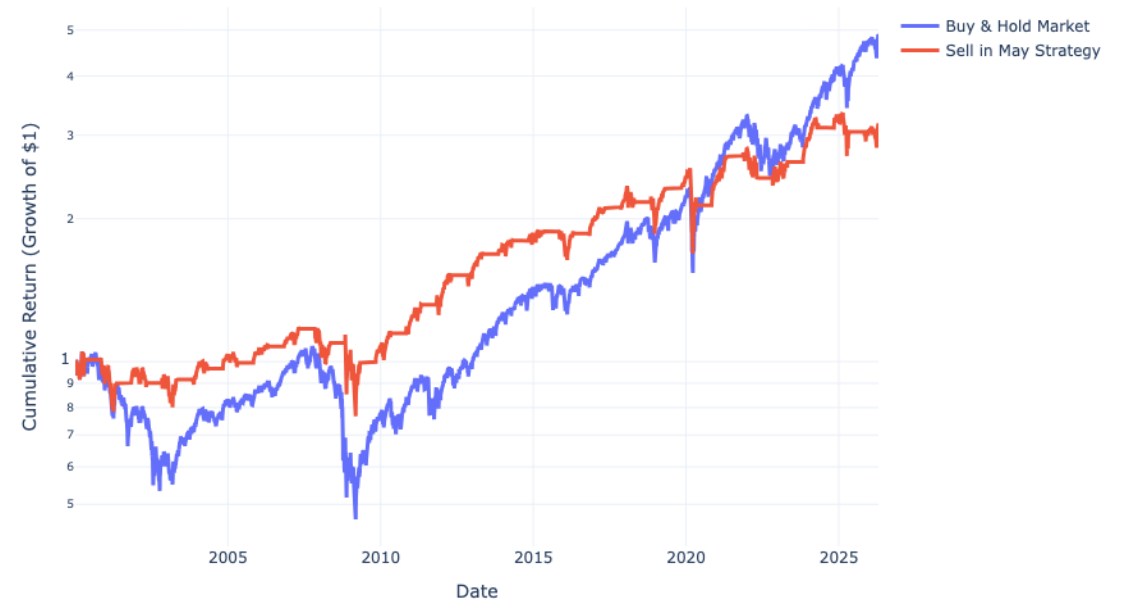
Go Short (-1) when:
Never

Go Long (+1) when:
Its November

Stay in Cash (0) when:
Its May

Strategy vs Market

Cumulative Return: Sell in May Strategy vs Buy & Hold Market



Moving Average Crossover Strategy

What It Is

A simple momentum-based strategy that uses two moving averages:

Fast MA: Short-term (e.g. 42-day)

Slow MA: Long-term (e.g. 252-day)

Core idea

Capture upward or downward trends by observing when the short-term average crosses the long-term average.

Moving Average Crossover Strategy

Go Long (+1) when:

Fast MA > Slow MA by a defined threshold (e.g. 3%)

Go Short (-1) when:

Fast MA < Slow MA by that same threshold

Stay in Cash (0) when:

Difference is within $\pm$ threshold

Threshold avoids noise and false signals.

Moving Average Crossover Strategy

Backtest Workflow

Calculate Fast and Slow MAs using rolling averages

Compute the difference between them

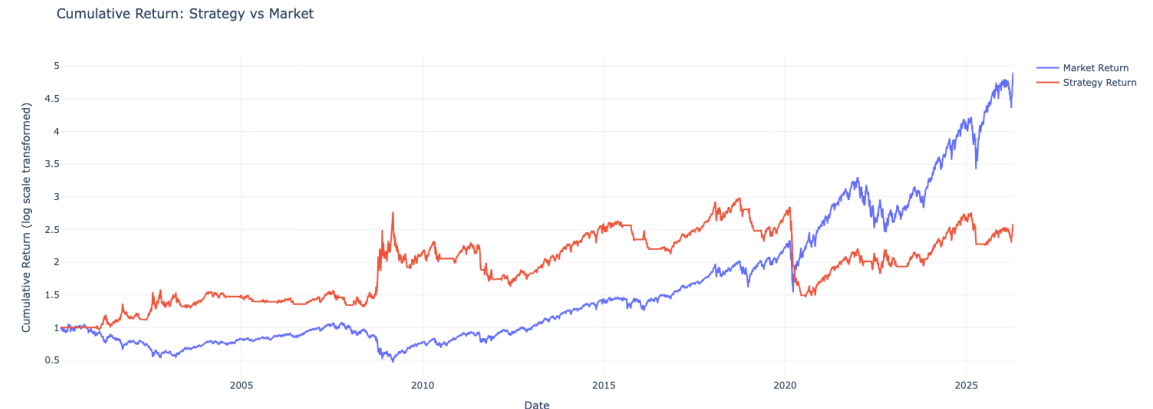
Generate regime signals using thresholds

Shift signals forward to simulate next-day execution

Multiply by daily market returns to calculate strategy returns

Evaluate using cumulative return and Sharpe ratio

Strategy vs Market



Moving Average Crossover Strategy

Strengths

Simple and interpretable

Works well in trending markets

Easy to implement and modify

Bollinger Bands Strategy

What It Is

A mean-reversion strategy based on volatility bands surrounding a moving average.

Middle Band: Typically a 20-day Simple Moving Average (SMA)

Upper Band: $\text{SMA} + 2 \times \text{rolling standard deviation}$

Lower Band: $\text{SMA} - 2 \times \text{rolling standard deviation}$

Core idea

Capture upward or downward trends by observing when the short-term average crosses the long-term average.

Bollinger Bands Strategy

Trading Logic

Go Long (+1) when:

Price crosses below the lower band (oversold)

Go Short (-1) when:

Price crosses above the upper band (overbought)

Exit Position (0) when:

Price returns to the middle band

Bollinger Bands Strategy

Backtest Workflow

Calculate Bollinger Bands: SMA and rolling standard deviation

Detect signal events:

Crosses below lower band → Long

Crosses above upper band → Short

Return to SMA → Close position

Shift signals forward to reflect real-world execution

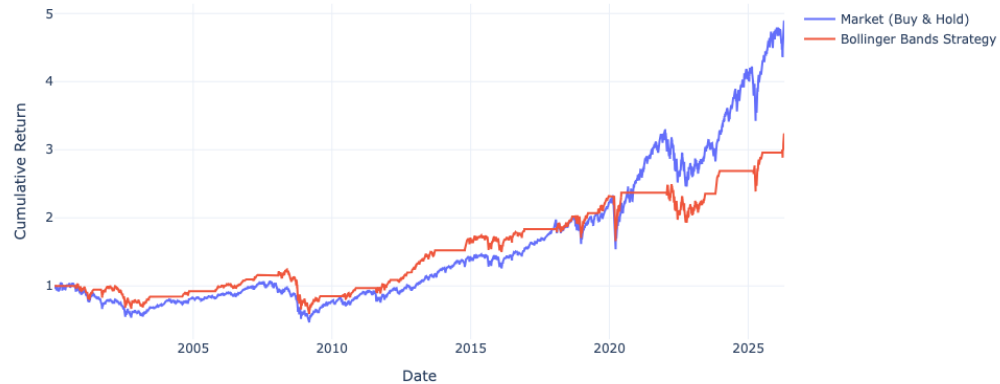
Multiply signals by market returns to get strategy returns

Evaluate performance via cumulative return, Sharpe ratio, and drawdown

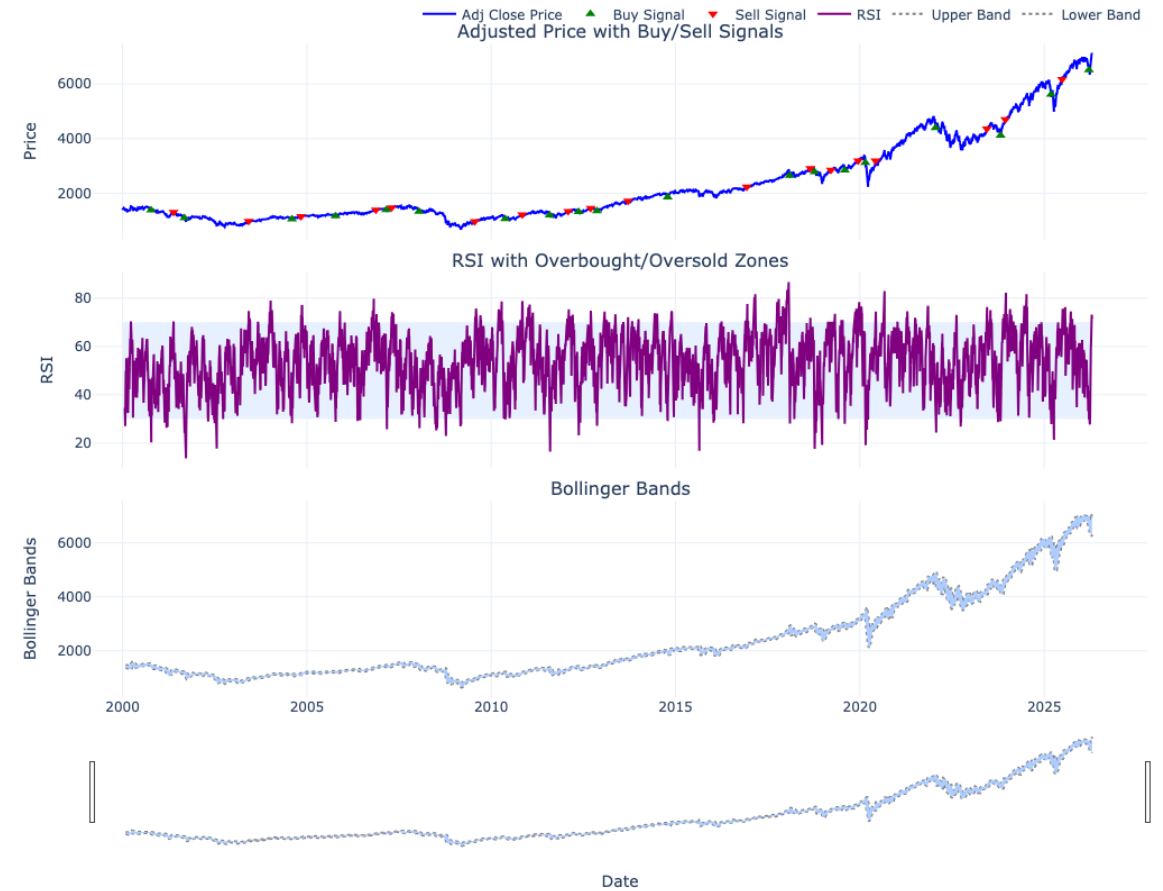
Bollinger Bands Strategy

Strategy vs Market

Cumulative Return: Strategy vs Market



Trading Strategy Visualization



Random

What It Is

Choose N positions /year at random

Core idea

Luck

Strategy vs Market

Cumulative Return: Random Strategy vs Buy & Hold Market

